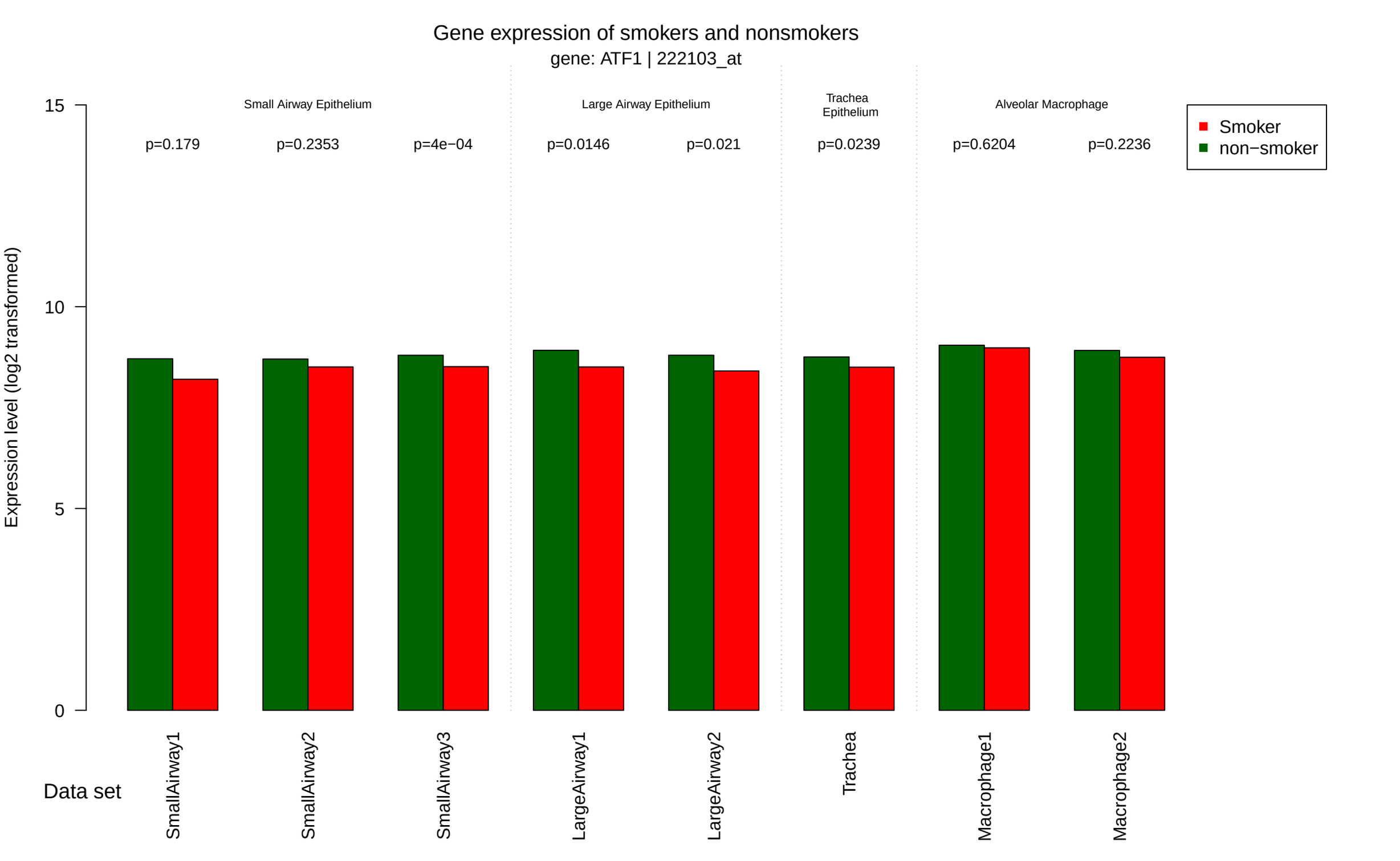
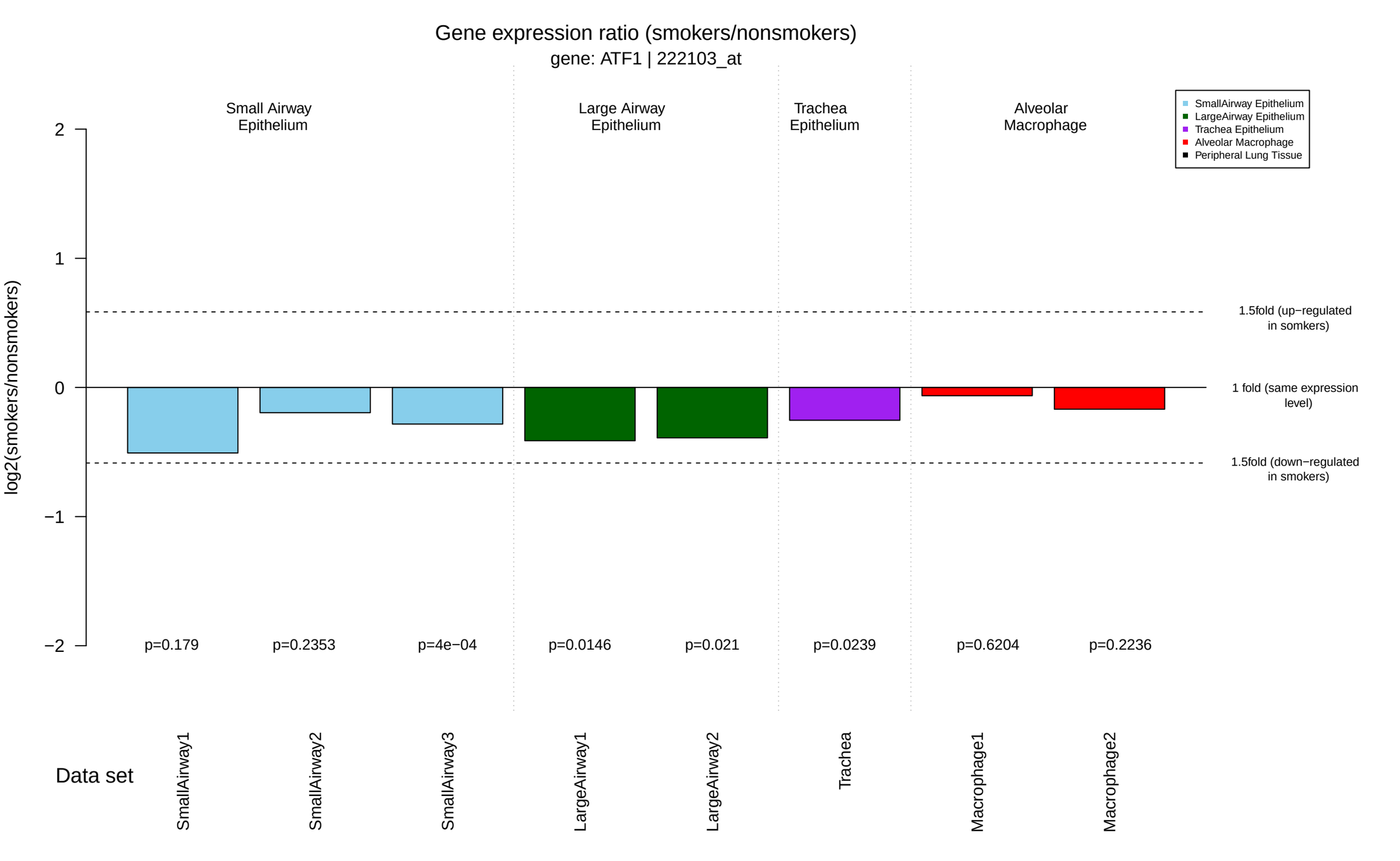
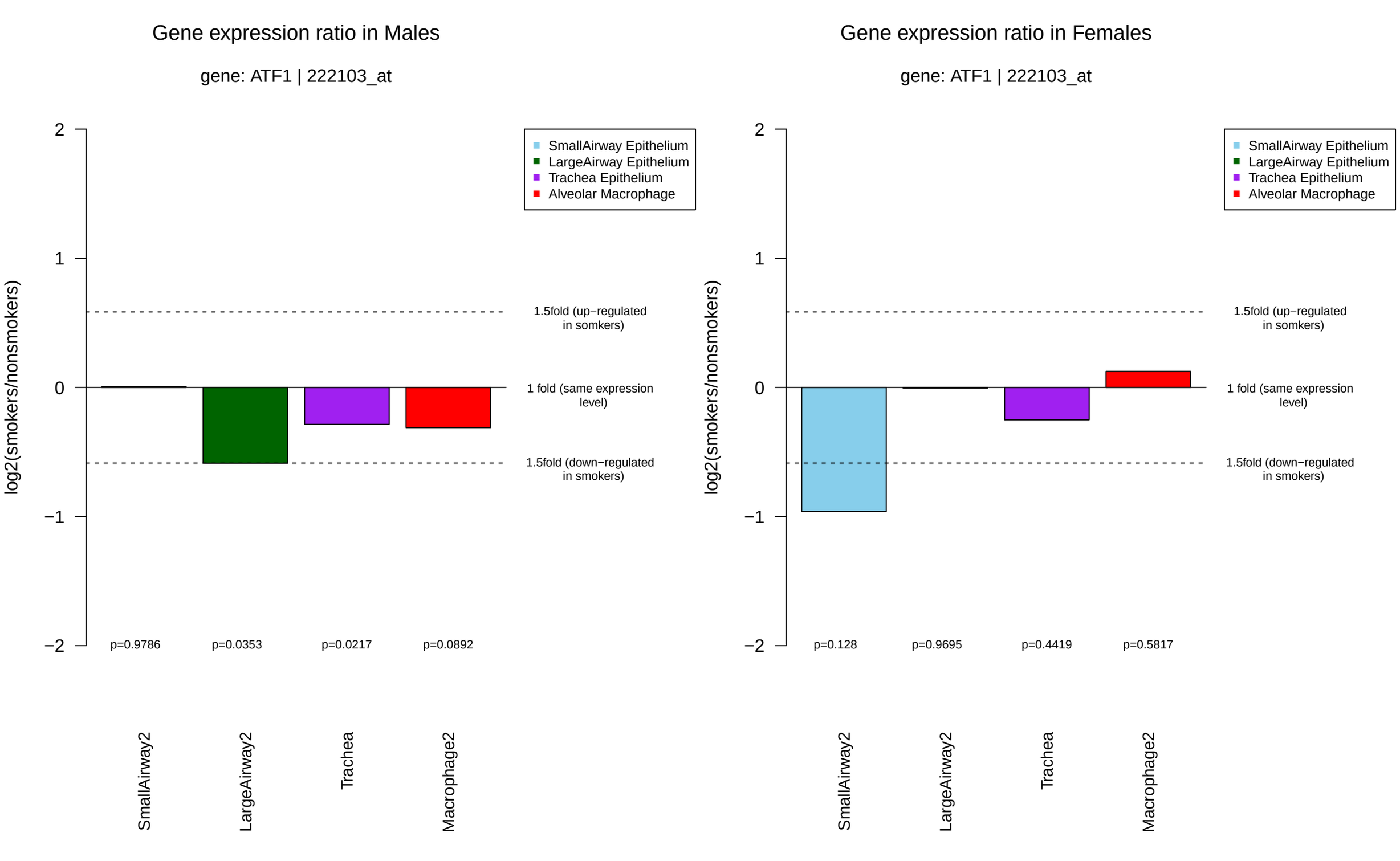
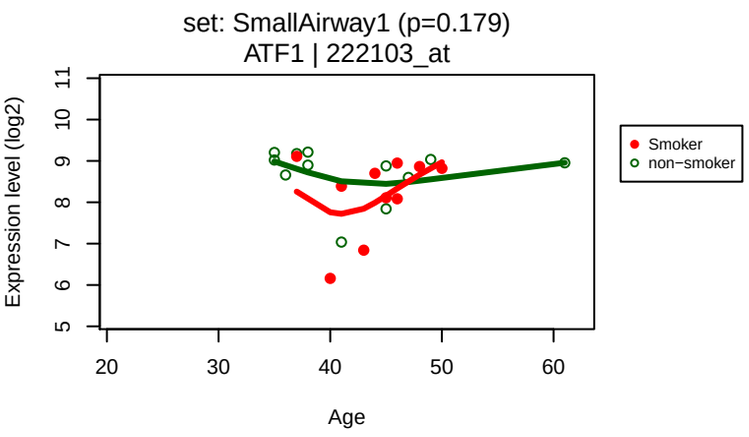
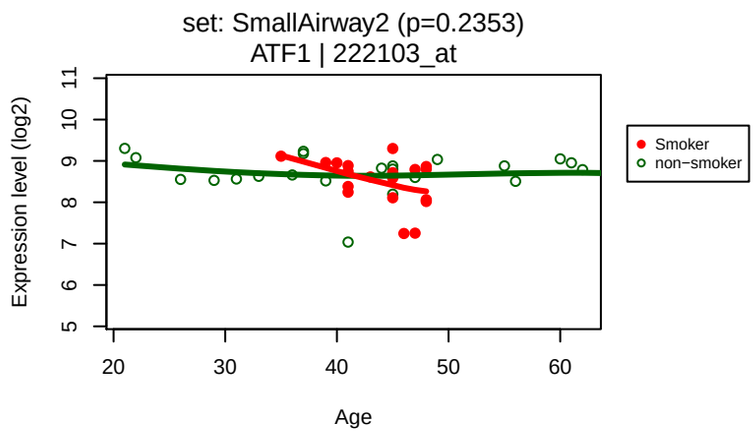
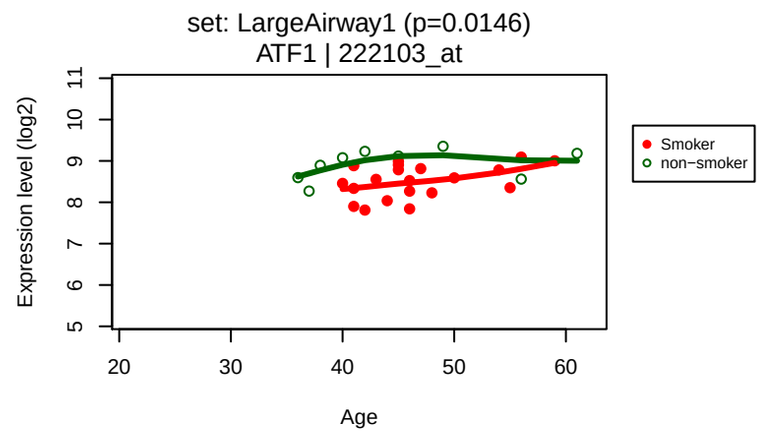

Gene expression of smokers and nonsmokers
gene: ATF1 | 222103_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.179
p=0.2353
p=4e−04
p=0.0146
p=0.021
p=0.0239
p=0.6204
p=0.2236
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

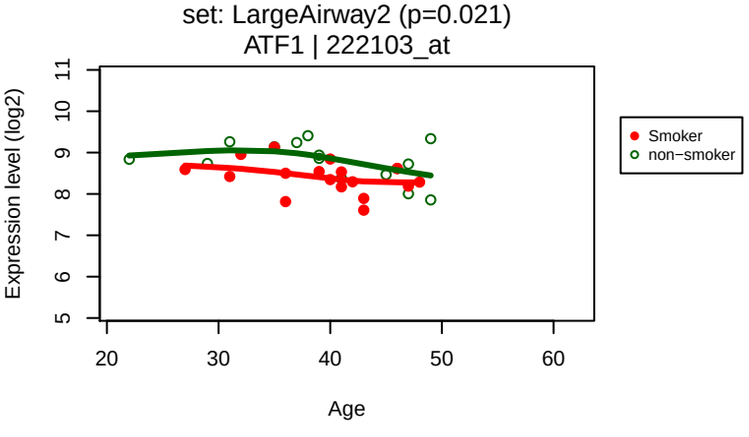

Gene expression ratio (smokers/nonsmokers)
gene: ATF1 | 222103_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.179
p=0.2353
p=4e−04
p=0.0146
p=0.021
p=0.0239
p=0.6204
p=0.2236
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

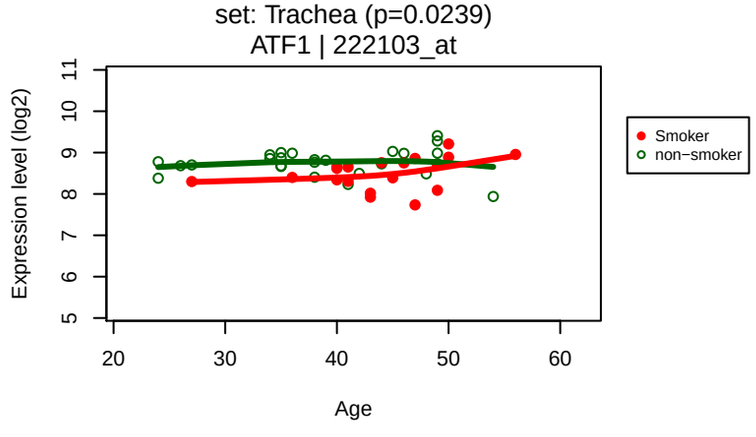

Gene expression ratio in Males
gene: ATF1 | 222103_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9786
p=0.0353
p=0.0217
p=0.0892
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ATF1 | 222103_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.128
p=0.9695
p=0.4419
p=0.5817
SmallAirway2
LargeAirway2
Trachea
Macrophage2

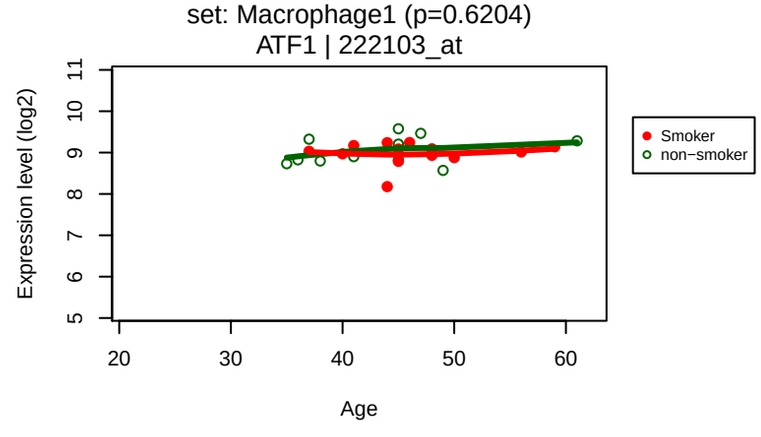

set: SmallAirway1 (p=0.179)
ATF1 | 222103_at
Expression level (log2)
Age
Smoker
non−smoker

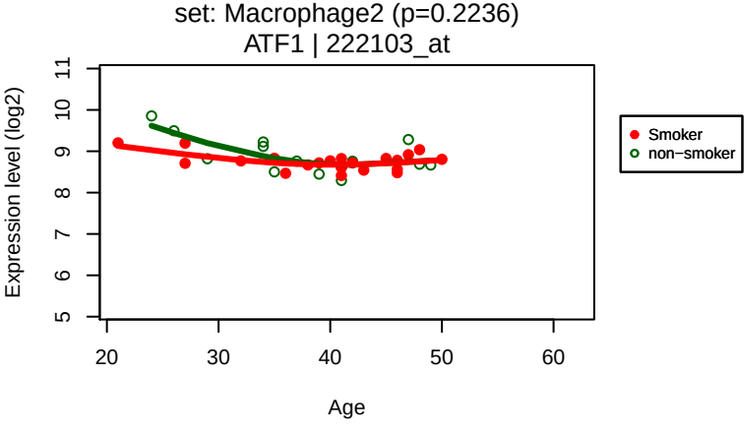

set: SmallAirway2 (p=0.2353)
ATF1 | 222103_at
Expression level (log2)
Age
Smoker
non−smoker

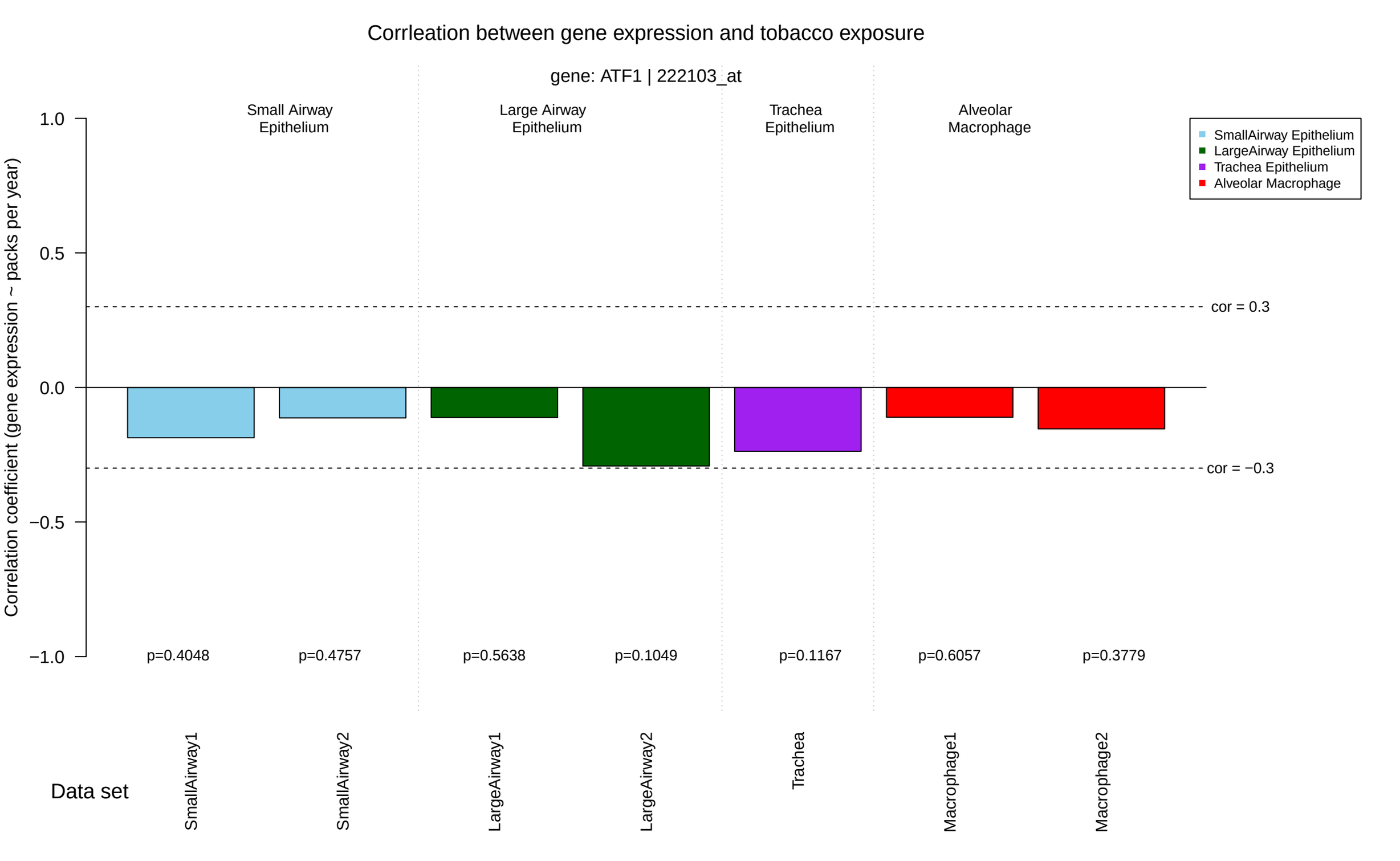

set: LargeAirway1 (p=0.0146)
ATF1 | 222103_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.021)
ATF1 | 222103_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0239)
ATF1 | 222103_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6204)
ATF1 | 222103_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2236)
ATF1 | 222103_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ATF1 | 222103_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.4048
p=0.4757
p=0.5638
p=0.1049
p=0.1167
p=0.6057
p=0.3779
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2